GCI WORLD 2026 SEPTEMBER — FINAL ASSIGNMENT | COMPANY A TELECOM POC [13]

From Reactive Churn Firefighting to Proactive Value Retention: A Risk-Ranked Lifecycle & Usage-Rescue PoC for Company A

Combining regularized gradient-boosted risk ranking (**M5 XGBoost**), pre-300-day handset lifecycle triggers, and Bayes prior-shift calibration to protect recurring contribution margin in a saturated mobile market.

1. EMPIRICAL AUDIT SCOPE [13]

100,000 Subscribers

100 merged columns (`Client.csv` ⋈ `Record.csv`); 49.56% case-control sample churn rate audited with zero data leakage (70/30 split, SEED=42).

2. PREDICTIVE & DECILE LIFT [10, 12]

0.7011 Test AUC

Up from 0.6938 tutorial baseline (+64.13% vs. 63.57% accuracy), cutting overfitting in half while achieving **81.0% sample precision** (32.5% re-based at 10% prior) in the Top-10% tier.

3. RE-BASED FINANCIAL IMPACT [8, 9]

+138,938 MU

Estimated annual Base Case net impact (+27,418 MU Conservative to +329,786 MU Optimistic) at Top-10% targeting after prior-shift re-basing and full offer costs.

Author: Islam Raad (AI Engineering, University of Mosul) | **Submission ID:** ACCOUNT_NAME

With Iraq Mobile Penetration at 100%–103%, Retention Beats Acquisition on Unit Economics [1, 2, 6]

48.1M (103%)
Iraq Active Mobile Lines [1]
Up from 46.0M (100.0%) in 2024; ITU reports 100.08/100 people [2]

427M (64%)
MENA Unique Subscribers [3]
>85%–90% prepaid mix in Iraq/Levant; widespread multi-SIM switching

+25% to +85%
Profit Lift per 5 pp Churn Cut [6]
Reichheld & Sasser (HBR): retention amortizes acquisition cost

Competitive Reality in Iraq & MENA Wireless Markets [1, 3, 4, 5]

Operator / Benchmark	Active Subscribers	Market Share / Scale	Strategic Retention Challenge
Zain Iraq (Zain Group) [4]	17.9M – 19.3M	~37% – 42% share USD 0.97B–1.1B rev	Protecting high-ARPU data & voice users against dual-SIM promo switching.
Asiacell (Ooredoo Group) [5]	17.69M – 19.13M	~38% – 40% share QAR 4,452M rev (24.8 QAR ARPU)	Retaining +8.1% YoY subscriber influx absorbed from displaced regional lines.
Korek Telecom [3, 5]	Contracting base	<15% share	Regulatory/interconnection disputes triggered mass SIM migration across Iraq.
Company A Dataset (PoC Target) [7, 13]	**100,000**	**58.72 MU Mean ARPU** 48.20 MU Median ARPU	**49.56% case-control sample churn** (real industry monthly/quarterly rate: **5%–10%** [7, 8]).

Why Blanket Retention Campaigns Fail in Telecom [7, 8]

- **Zero Greenfield Headroom:** At 100.08 subscriptions per 100 people in Iraq [2], every "new" gross addition is either a secondary SIM or poached from a rival at high dealer commission and onboarding cost.
- **Untargeted Discounts Destroy Margin:** Offering a 10 MU monthly discount to all 100,000 subscribers costs 1,000,000 MU per wave—mostly subsidizing loyal customers who would have stayed anyway [8].
- **Timing Beat Generosity:** Once a subscriber zeroes out voice minutes (`mou_Mean == 0`) or swaps primary SIMs, churn reaches **79.81%** [13]—interventions must fire *before* disengagement completes.
- **Unit Currency Convention:** Because Company A's dataset is denominated in North American wireless billing scale (mean `rev_Mean` = 58.72 MU vs. ~6.80 USD in Iraq [5, 13]), all PoC financials are reported in **Dataset Monetary Units (MU)**.

So What / Decision Implication: In a 100%+ penetrated market [1, 2], EBITDA growth depends on surgical retention—identifying the **Top-10% highest-risk, positive-margin subscribers** early enough to intervene profitably [7, 8].

Sources: [1] DataReportal Iraq 2025; [2] World Bank/ITU WDI; [3] GSMA MENA 2024; [4] Zain Group AR; [5] Ooredoo AR; [6] Reichheld & Sasser (1990); [7] Neslin et al. (2006).

100,000-Subscriber Audit Reveals Informative Missingness, 10 Duplicate Pairs & Unit Anomalies [13]

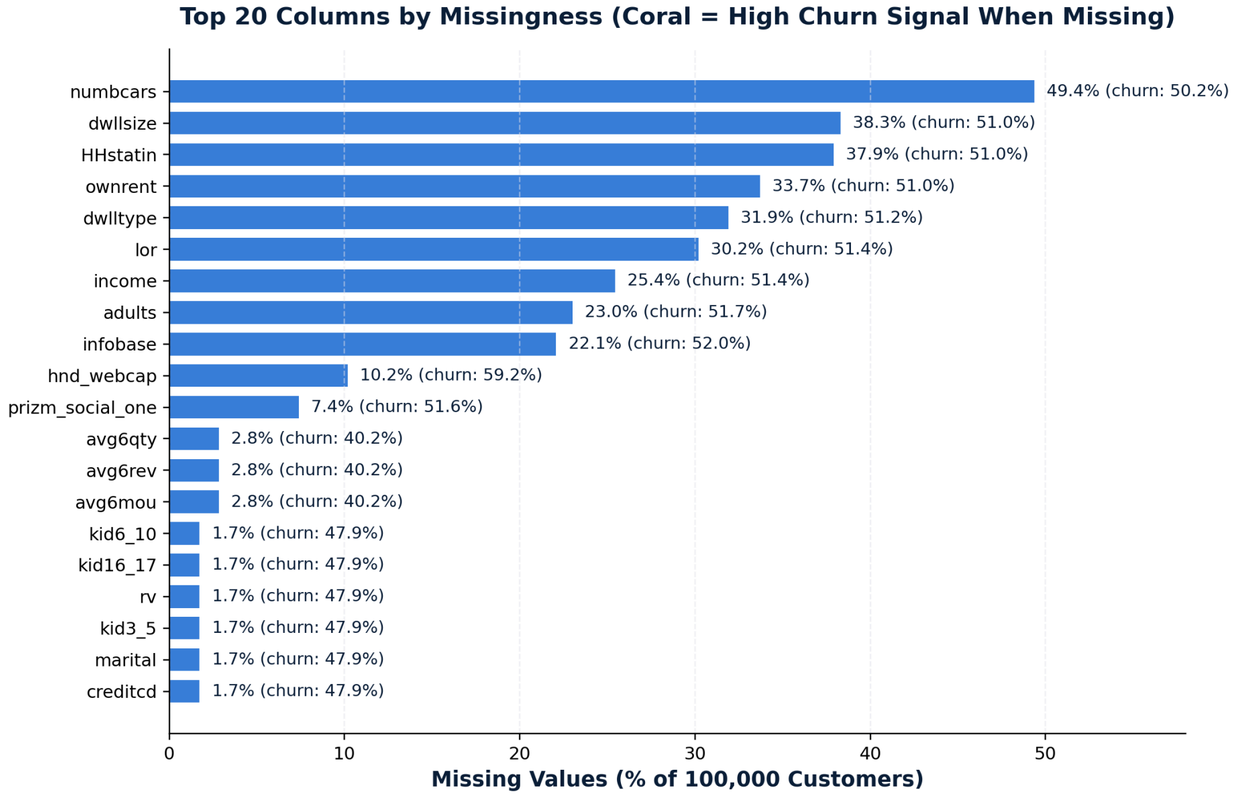


Figure 1: Top-20 columns by missingness (%) on `Client.csv` ⋈ `Record.csv` (N = 100,000) with churn-rate delta when missing vs. present [13].

100,000 × 100

Exact 1:1 Inner Join [13]

51 Record + 50 Client cols; 0 duplicate IDs (`1,000,001–1,100,000`)

76.09%

Churn When `change_mou` Missing

n = 891 (+26.77 pp vs. present); `hnd_webcap` NaN = 59.15% churn

10 Cols Dropped

Near-Duplicate Pairs (|ρ| > 0.995)

Dropping 10 redundant cols improves 5-fold CV AUC (0.6940 → 0.6947)

Critical Audit Findings & Documentation Discrepancies Fixed [13]

Audit Dimension	Empirical Finding in Dataset	Pipeline Engineering Action
1. `change_mou` & `change_rev` Units	Overview docx calls them "% change", but **22.79%** of `change_mou` < -100 (min -3,875) and r = **0.9025** with `mou_Mean - avg3mou` (raw difference).	Construct true relative ratios `mou_ratio_3m` and `rev_ratio_3m` alongside raw deltas.
2. Negative `eqpdays` Values	**133 rows** have `eqpdays < 0` (min -5; 34 rows = -1, which causes `np.log1p` `RuntimeWarning` in `tutorial.ipynb`).	Add `eqpdays_negative_flag` and clamp `eqpdays_clean = max(eqpdays, 0)` before log/ratio transforms.
3. Correlated Missing Blocks	**1,732 rows (1.73%)** miss all 11 demographic fields (`truck` .. `kid16_17` , 47.86% churn); **2,839 rows** miss `avg6*` (tenure < 6m, 40.23% churn).	Create explicit missingness indicators (`missing_change_mou` , `missing_hnd_webcap` , `missing_avg6` , `missing_demo_block`).
4. 20 Collinear Pairs (|ρ| > 0.98)	`totcalls` / `adjqty` (ρ=0.9995), `totmou` / `adjmou` (ρ=0.9995), `plcd_vce` / `attempt` (ρ=0.9994), `comp_vce` / `complete` (ρ=0.9992).	Drop 10 redundant duplicate columns to stabilize split gain and permutation importance.

So What / Decision Implication: Missing values in telecom billing logs are **behavioral signals, not random noise**—preserving missingness indicators and fixing `change_mou` scaling directly lifts ranking accuracy [13].

Sources: [13] GCI Company A Dataset (`Client.csv` , `Record.csv`) & Overview Docx Audit (`outputs/tables/missing.csv` , `docs/data_doc_discrepancies.md`).

Churn Surges +23.83 pp After Handset Age Crosses 300 Days (36.85% → 60.69%) [13]

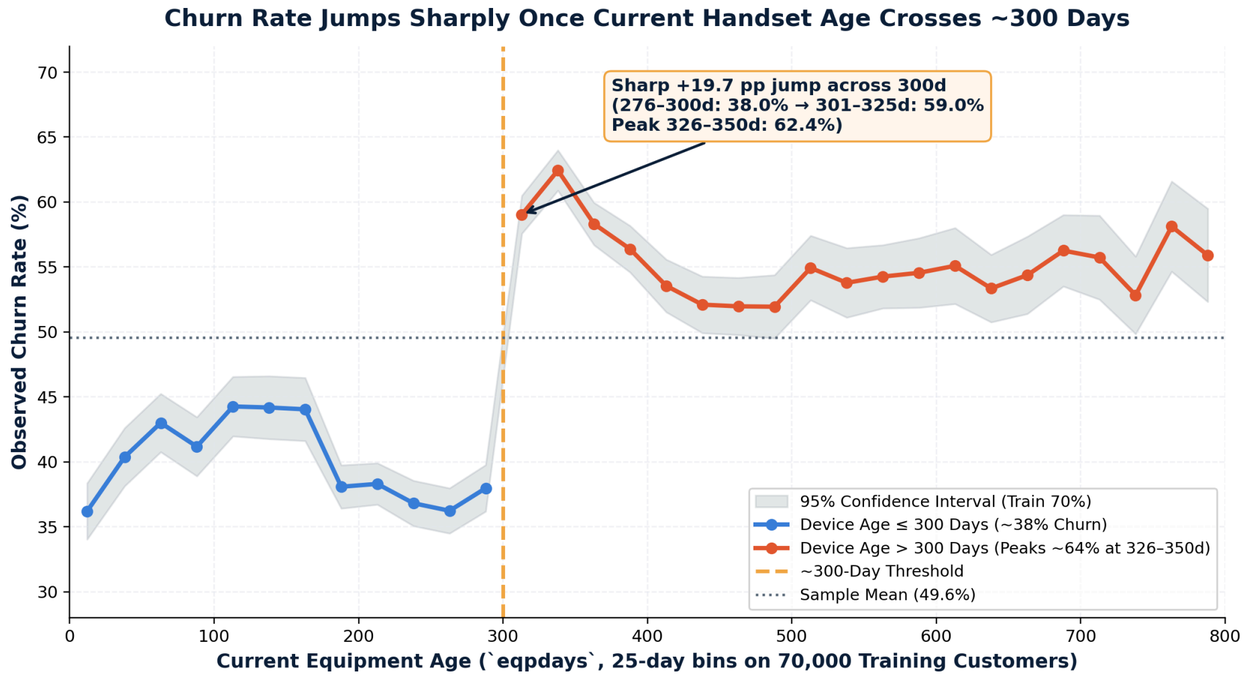


Figure 2: Churn rate across 25-day handset age (`eqpdays`) bins on the 70% training split (n = 70,000) with 95% Wilson confidence intervals [13].

36.85%
Pre-Threshold Churn (260–300d)
n = 4,792 train accounts (36.87% full dataset)

60.69%
Post-Threshold Churn (301–340d)
n = 6,774 train accounts; peaks at 62.44% in 326–350d

+23.83 pp
Verified Jump Across 300d
95% Boot CI: [+22.12, +25.62] pp | RD Jump: +23.88 pp

Subgroup Robustness Audit Around the 300-Day Boundary (Train Split) [13]

Customer Subgroup	260–300d Churn (n)	301–340d Churn (n)	Jump (Δ pp)
All Training Accounts	36.85% (4,792)	60.69% (6,774)	**+23.83 pp**
Single-Handset Accounts (`phones == 1`)	29.10% (2,869)	63.18% (4,932)	**+34.08 pp**
New Handsets Only (`refurb_new == 'N'`)	34.23% (4,128)	60.65% (6,087)	**+26.42 pp**
Tenure ≥ 10 Months Only (`months ≥ 10`)	41.23% (3,090)	60.69% (6,774)	**+19.46 pp**
Prime Credit Tier (`crclscod == 'AA'`)	34.11% (1,897)	63.42% (3,152)	**+29.31 pp**
Multi-Handset Accounts (`phones ≥ 2`)	48.41% (1,923)	54.02% (1,842)	**+5.60 pp**

So What / Decision Implication: `eqpdays` is the strongest single predictor in the dataset (univariate AUC = 0.5751), and the **240–300 day pre-spike window** provides a clear 60-day operational runway for proactive retention [13].

Sources: [13] Company A Dataset 70% Train Split (`outputs/tables/eqpdays_bins25.csv` , `outputs/tables/threshold_robustness.csv`).

The 300-Day Spike Coincides with the Months 10–12 Renewal Cliff, Yet Device Age Persists Later [13]

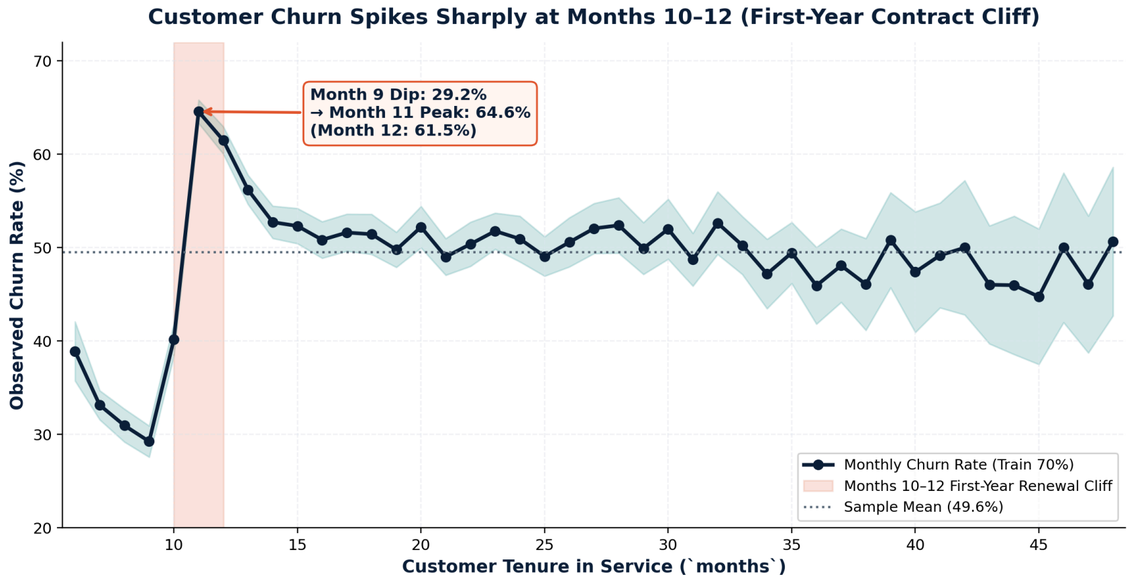


Figure 3: Churn rate by account tenure (`months` 6–36) showing the dramatic doubling from Month 9 (29.25%) to Month 11 (64.57%) [13].

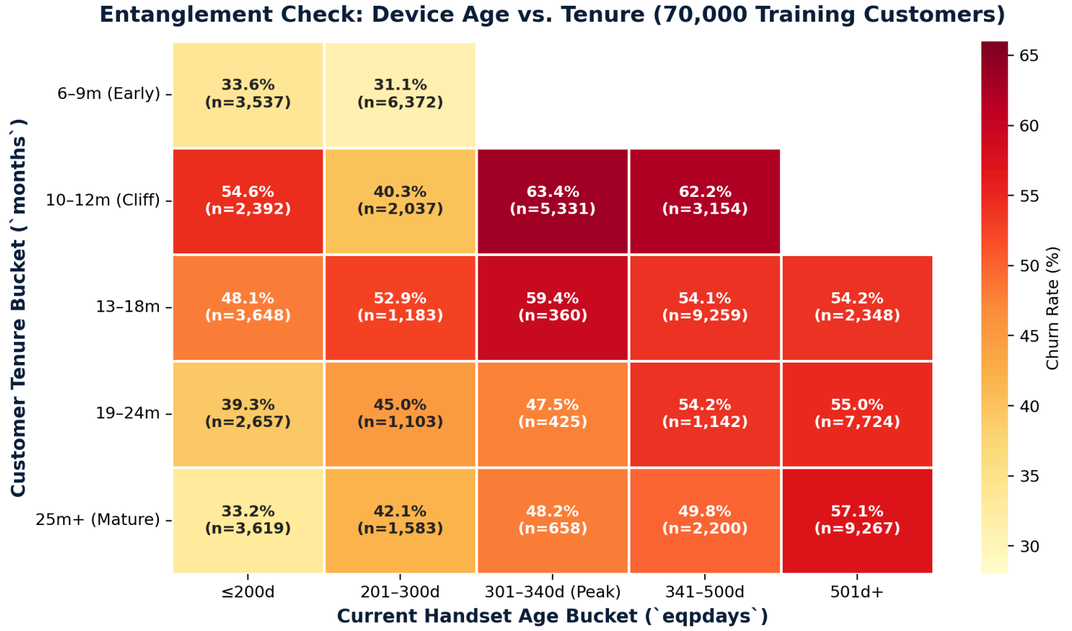


Figure 4: 2D Churn Rate (%) & Sample Size across Tenure (`months`) × Handset Age (`eqpdays`) on the 70% training split [13].

Why Disentangling `months` and `eqpdays` Matters for Intervention Design [13]

- **First-Year Coupling:** On a subscriber's first handset (`phones == 1`), `eqpdays ≈ 30 × months` . Exactly **0 customers** have `months ≤ 9` and `eqpdays > 300` (max `eqpdays` for `months ≤ 9` is **288 days**). Thus, crossing 300 days on a first phone is also crossing into the **Month 10–12 contract renewal cliff** (29.25% → 64.57%).
- **Independent Device-Age Effect in Mature Cohorts:** Holding tenure fixed above 12 months, subscribers with a newer/upgraded handset (`≤200d`) churn far less than peers on aging handsets (`501d+`): **39.2% vs. 55.0%** in Months 19–24 (**-15.7 pp**) and **33.2% vs. 57.1%** in Months 25+ (**-23.8 pp**).

So What / Decision Implication: Because first-year contract renewal and handset aging co-move, our **Lane 2 intervention pairs a Month 8–9 renewal check with a handset trade-in voucher**, verified via an 80/20 holdout pilot [13].

Sources: [13] Company A Dataset 70% Train Split (`outputs/tables/tenure_months_churn.csv` , `outputs/tables/tenure_device_entanglement.csv`).

3. EXPLORATORY DATA ANALYSIS — FINDING 3

Zero-MOU Accounts Churn at 79.81% on Low ARPU (23.11 MU), Requiring Value-Gated Triage [13]

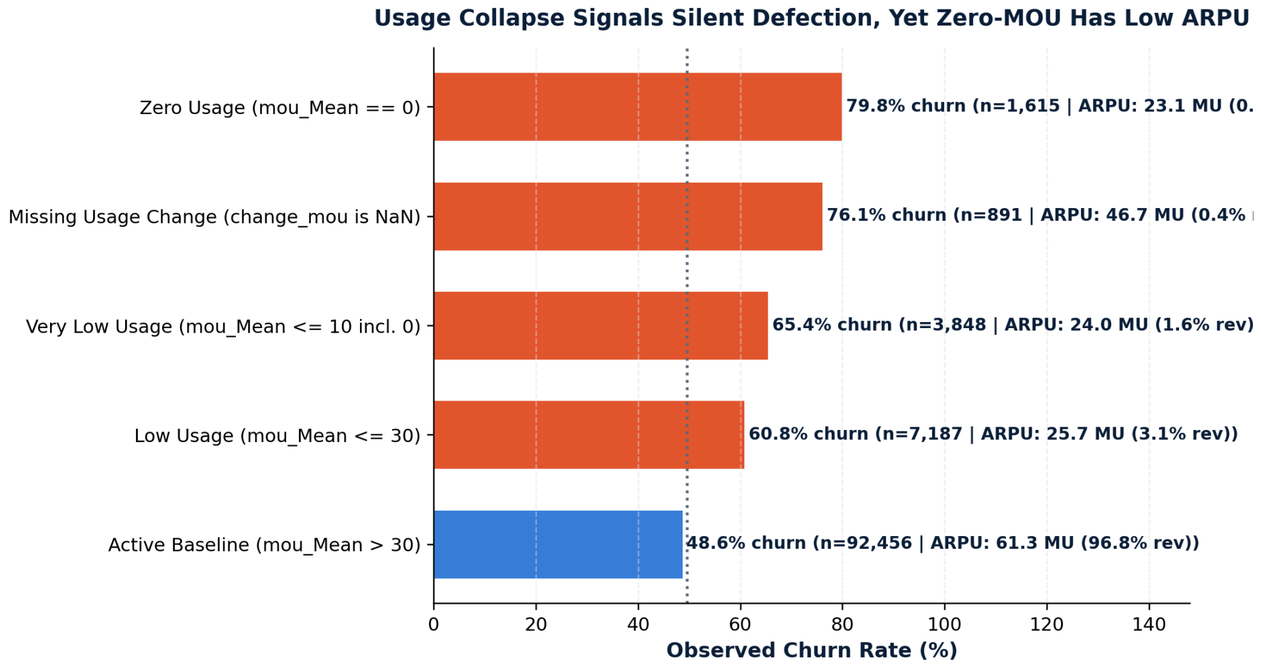


Figure 5: Churn rate (%) vs. Mean Monthly Revenue (`rev_Mean`, MU) across voice usage segments [13].

Three Commercial Traps Uncovered by Segment & Fairness Audits [13]

- **Trap 1 — The Dormant SIM Trap (`mou_Mean == 0`):** The 1,615 zero-usage accounts churn at **79.81%** (78.48% train), making them easy for any ML model to flag. Yet their mean ARPU is only **23.11 MU** (median 17.14 MU, **0.64% of total revenue**). Spending 7.50–10.00 MU on human call-center outreach for already-silent SIMs destroys campaign ROI.
- **Trap 2 — Mean ARPU Alone Is Blind (`rev_Mean`):** Overall monthly revenue is virtually identical between churners (**58.21 MU** mean / 47.49 MU median) and stayers (**59.22 MU** mean / 48.88 MU median; r = -0.0109). What predicts churn is *relative usage trajectory* (`change_mou`, `mou_ratio_3m`) and *overage friction* (`ovrrev_Mean > 20 MU` → 52.05% churn).
- **Trap 3 — Demographic Disparity (`ethnic`):** Raw churn across `ethnic` codes spans **18.37 pp** (Code `0`: **58.08%** vs. Code `Z`: **39.71%**). To prevent discriminatory offer allocation, we **exclude** `ethnic` from production scoring—verified to cost only **-0.0016 validation AUC** (0.7073 → 0.7057).

79.81%
Zero-MOU Churn (n=1,615)
ARPU = 23.11 MU → Automated SMS only

58.21 vs 59.22
Churned vs Stayed Mean MU
Static ARPU cannot separate risk

-0.0016 AUC
Cost of Dropping `ethnic`
Fairness achieved with ~zero accuracy loss

So What / Decision Implication: High churn probability alone is not enough to trigger costly human outreach—offers must be gated by **active value (`rev_Mean ≥ 30 MU`, `mou_Mean > 0`)** and **fairness-audited features** [8, 13].

Sources: [13] Company A Dataset (`outputs/tables/usage_collapse_segments.csv`, `outputs/tables/ethnic_fairness_audit.csv`).

We Frame Retention as a Leakage-Safe Top-k% Risk-Ranking Engine Linked to Two Action Lanes [8, 12]

End-to-End Leakage-Safe Analytical & Decision Workflow (SEED = 42)

Every transformation, encoder, and threshold is fit strictly on the 70% training split; the 30% test split is touched once for final reporting.

1. Data Audit & 1:1 Join
- **Client.csv + Record.csv**
  100,000 rows × 100 columns
- **Stratified 70/30 Split**
  70,000 Train / 30,000 Test
- **Data Hygiene Fixes**
  Clamp eqpdays < 0 (133 rows)
  Drop 10 duplicate _Mean cols

2. 20 Domain Features
- **Lifecycle & Device Signals**
  dev_age_gt_300, tenure_10_12
  eqp_per_tenure_day ratio
- **Usage Slump & Friction**
  mou_ratio_3m, care_intensity
- **Fairness Guardrail**
  Exclude ethnic (107 fair cols)

3. 5-Fold Stratified CV
- **7 Candidate Models**
  Dummy, LogReg, Shallow Tree, XGBoost (M3–M5), LightGBM (M6)
- **Overfitting Control**
  depth=5, subsample=0.8, colsample=0.8, reg_lambda=2
- **Primary Metric: ROC-AUC**

4. Top-k% Profit Engine
- **Rank by Risk Score $\hat{p}_i$**
  Select Top-10% (10,000 accts)
- **Prior-Shift Re-basing**
  Adjust 49.6% sample prior to realistic 5%–10% base rate
- **Two-Lane Intervention**
  Matched to root risk driver

Why Probability Ranking Beats 0.5-Cutoff Classification:
A flat 0.5 cutoff flags ~15,000 test accounts regardless of campaign budget. Ranking by $\hat{p}_i$ lets CRM size contact volume k% to maximize net saved CLV minus offer cost.

Candidate Problem Formulation Decision Matrix (Evaluated Before Modeling) [7, 8, 13]

Candidate Problem Framing	Primary Target & Action	Data Support (30%)	Intervenability (25%)	Net Impact (25%)	Causal Rigor (20%)	Weighted Score	Strategic Role in Proposal
P1. Proactive Top-10% Risk-Ranked Retention Engine	Rank all 100k users by $\hat{p}_i$; target Top-10% (`rev_Mean ≥ 30 MU`)	5.0 / 5	4.5 / 5	5.0 / 5	4.5 / 5	**4.78 / 5.00**	**Primary Core Architecture (Lane 1 + Scoring Engine)**
P2. Pre-300-Day Handset Refresh & Month 8–9 Renewal Pilot	Target `eqpdays` in [240, 300]d (`n = 8,815` , ARPU 66.91 MU)	4.5 / 5	4.5 / 5	4.0 / 5	4.5 / 5	**4.38 / 5.00**	**Integrated as Dedicated Lifecycle Track (Lane 2 Pilot)**
P3. Dormant / Zero-MOU Account Reactivation Triage	Target `mou_Mean ≤ 10` (`n = 3,848` , ARPU 24.00 MU)	4.5 / 5	2.5 / 5	2.5 / 5	4.0 / 5	**3.40 / 5.00**	**Automated Low-Cost Digital/SMS Guardrail Only**

So What / Decision Implication: Combining **P1 (Top-10% ML Risk Ranking)** with **P2 (Pre-300-Day Lifecycle Trigger)** and a **P3 Dormant-Account Cost Filter** maximizes protected CLV per MU of retention spend [8].

Sources: [7] Neslin et al. (2006); [8] Verbeke et al. (2012); [12] Pedregosa et al. (2011); `docs/problem_decision_matrix.md`.

Solving the Case-Control Trap: Re-Basing 49.56% Sample Churn to Realistic 5%–10% Priors [7, 9]

Why Unadjusted Sample Precision Overstates Production ROI by 3.3× [7, 9]

- **The Case-Control Calibration Structure:** Company A's dataset has a **49.56% churn rate** (`49,562` churned vs. `50,438` stayed) [13]. As documented by Neslin et al. (2006) on the Teradata/Duke wireless benchmark [7] and highlighted in the GCI Tutorial PDF, this is a **50/50 case-control sample** oversampled from a real operator environment where monthly/quarterly churn is **5% to 10%** [3, 7, 8].
- **Ranking Is Invariant, Precision Is Not:** Under prior probability shift (where class-conditional feature distributions P(X | Y) remain invariant), a model's **ROC-AUC is unchanged**, but the fraction of true churners inside the top-k% tier depends directly on the true base rate π_r [9].
- **Exact Bayes Prior-Shift Formula (Saerens et al., 2002) [9]:** Given sample prior $\pi_s = 0.4956$, sample top-k% precision P_s, and real production prior $\pi_r \in \{0.05, 0.10\}$:

$$P_{real}(k) = [\, P_s(k) \cdot (\pi_r / \pi_s) \,] \;/\; [\, P_s(k) \cdot (\pi_r / \pi_s) + (1 - P_s(k)) \cdot ((1 - \pi_r) / (1 - \pi_s)) \,]$$

At **Top-10%** (n = 10,000), sample precision P_s = 81.00% (1.63× lift) maps to **P_{real} = 32.53% (3.25× lift) at π_r = 10%** and **P_{real} = 18.59% (3.72× lift) at π_r = 5%** [9].

Side-by-Side Impact of Prior-Shift Correction at Top-10% (n = 10,000) [9]

Metric at Top-10% Tier	Unadjusted Sample (π_s = 49.56%)	Re-Based Base Case (π_r = 10.0%)	Re-Based Conservative (π_r = 5.0%)
Targeted Subscribers (n)	10,000	10,000	10,000
True Precision in Top-10%	81.00%	**32.53%**	**18.59%**
True Churners Reached	8,100	**3,253**	**1,859**
False Positives Contacted (Incur full offer cost c)	1,900	6,747	8,141
Lift Over Random Outreach	1.63×	**3.25×**	**3.72×**
Estimated Net Impact (MU) (After paying c on all 10,000)	+457,708 MU (3.3× inflated)	**+138,938 MU** (Realistic Base)	**+27,418 MU** (Realistic Low)

So What / Decision Implication: Reporting **prior-shifted precision (32.5% at 10% base rate; 18.6% at 5%)** prevents CFO budget over-commitment while proving the model still delivers a **3.25×–3.72× concentration of churners** [9].

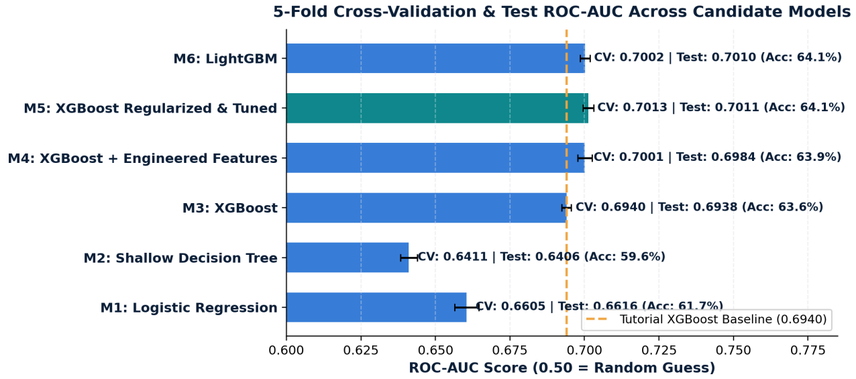
Sources: [7] Neslin et al. (2006); [8] Verbeke et al. (2012); [9] Saerens, Latinne, & Decaestecker (2002), Neural Computation.

Final Model M5 (Regularized XGBoost) Reaches 0.7013 CV / 0.7011 Test ROC-AUC & Cuts Overfitting in Half [10, 12]

Systematic 7-Model Progression on 70% Training Split (5-Fold Stratified CV) & Held-Out 30% Test Split (SEED = 42) [10, 11, 12]

Selected Model: XGBoost Regularized & Tuned (M5) | Metric: ROC-AUC = 0.7011 (Test) / 0.7013 (CV) | Accuracy = 0.6413

ID	Model Name & Configuration	Features	5-Fold CV ROC-AUC	Test ROC-AUC	Test Accuracy	Train Accuracy	Train–Test Gap	Test Top-10% Precision
M0	**Dummy Majority Baseline** Predicts majority class (stayed=0)	98	**0.5000** ±0.0000	**0.5000**	**0.5044** (50.44%)	50.44%	0.00 pp	**49.6%**
M1	**Logistic Regression (Pipeline)** Linear baseline (impute + scale + OHE, 107 fair features)	107	**0.6605** ±0.0041	**0.6616**	**0.6172** (61.72%)	61.92%	0.21 pp	**72.9%**
M2	**Shallow Decision Tree (depth=4)** Rule-based explanation model (max_depth=4, 107 fair features)	107	**0.6411** ±0.0028	**0.6406**	**0.5956** (59.56%)	59.77%	0.21 pp	**73.1%**
M3	**XGBoost (Tutorial Baseline)** TUTORIAL BASELINE 98 raw features, LabelEncoder, depth=6, lr=0.05, n=300	98	**0.6940** ±0.0016	**0.6938**	**0.6357** (63.57%)	73.49%	9.92 pp	**79.2%**
M4	**XGBoost + Engineered Features** 107 fair features (20 engineered, -10 dups, no ethnic), depth=6, lr=0.05, n=300	107	**0.7001** ±0.0024	**0.6984**	**0.6386** (63.86%)	74.00%	10.14 pp	**80.7%**
M5	**XGBoost Regularized & Tuned** FINAL SELECTED 107 fair features, depth=5, lr=0.03, n=550, subsample=0.8, colsample=0.8, min_child_weight=20	107	**0.7013** ±0.0018	**0.7011**	**0.6413** (64.13%)	69.36%	5.23 pp	**81.0%**
M6	**LightGBM (Gradient Boosted Trees)** 107 fair features, num_leaves=31, lr=0.03, n=500, subsample=0.8	107	**0.7002** ±0.0017	**0.7010**	**0.6409** (64.09%)	71.17%	7.07 pp	**81.7%**



Why M5 (Regularized XGBoost) Is Selected Over M3 & M6 [10, 11]

- **Beats Tutorial Baseline (M3) Across Every Metric:** Test ROC-AUC rises from **0.6938 → 0.7011** (+0.0073), 5-fold CV ROC-AUC rises from **0.6940 → 0.7013**, Test Accuracy rises from **0.6357 (63.57%) → 0.6413 (64.13%)**, and Top-10% sample precision jumps from **79.2% → 81.0%** (+183 more true churners caught per 10,000 targeted).
- **Cuts Overfitting Gap Nearly in Half:** Tutorial M3 (`max_depth=6` , unregularized) overfits heavily (Train Acc 73.49% vs. Test Acc 63.57% = **9.92 pp gap**). Regularizing M5 (`max_depth=5` , `min_child_weight=20` , `subsample=0.8` , `colsample_bytree=0.8` , `reg_lambda=2.0`) shrinks the gap to **5.23 pp** (`Brier score = 0.2185`).
- **Cross-Checked Against LightGBM (M6):** LightGBM (`0.7002` CV AUC, `0.7010` Test AUC, `0.6409` Test Accuracy) confirms the ~0.701 AUC ceiling on these 107 fair tabular features.

So What / Decision Implication: Domain feature engineering (+20 lifecycle/usage features, -10 duplicates, excluding `ethnic`) plus tree regularization lifts Top-10% precision to **81.0%** while halving overfitting [10, 12].

Sources: [10] Chen & Guestrin (2016) XGBoost; [11] Ke et al. (2017) LightGBM; [12] Pedregosa et al. (2011) Scikit-learn (`outputs/tables/experiment_log.csv`).

5. MODEL EXPLAINABILITY & ACTIONABILITY

How the Model Ranks Risk: Handset Lifecycle, 3-Month Usage Slumps & Plan Fit Drive Predictions [10, 13]

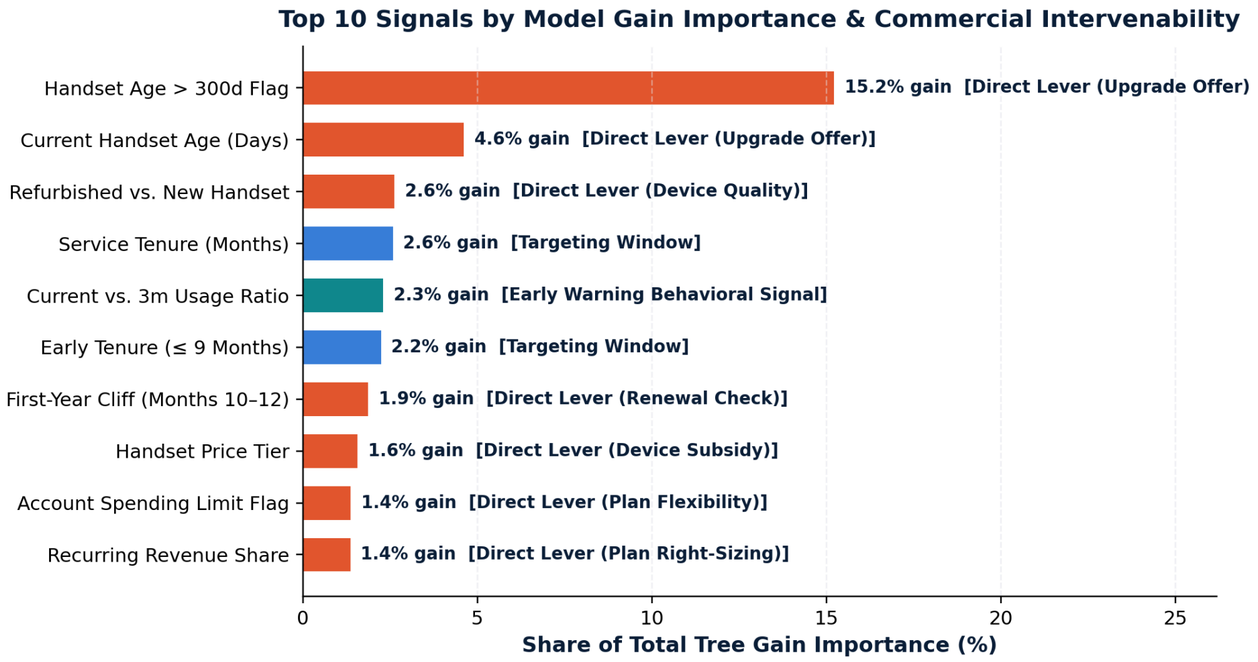


Figure 6: Top-10 features in Final Model M5 by tree split gain share (%), color-coded by commercial intervenability [10, 13].

Plain-Language Intuition: How Gradient-Boosted Trees Score Individual Churn Risk

Instead of relying on a single rigid rule, 550 shallow decision trees sequentially add or subtract risk points based on lifecycle, usage, and care signals

Tree Group A: Device & Tenure

Handset Age > 300 Days?

- No (≤300d) **-0.14 risk**
- Yes (>300d) **+0.21 risk**

- Also checks Months 10–12 cliff
- Also checks Refurbished phone

→ **Direct Upgrade / Renewal Lever**

+

Tree Group B: Usage Slump

Minutes Dropping vs 3m Avg?

- Stable/Up **-0.08 risk**
- Slump >30% **+0.16 risk**

- Checks change_mou & ratio_3m
- Flags zero-MOU dormant SIMs

→ **Early-Warning Rescue Trigger**

+

Tree Group C: Plan Fit

Overage & Care Calls?

High overage share + no spending limit (ASL=N) **+0.09 risk**

- Checks totmrc & ovrrev
- Checks custcare_Mean

→ **Plan Right-Sizing Lever**

→

Calibrated Score p̂_i

- **Top 10% Risk Tier** 81.0% Sample Churn
- **Middle 80% Tier** Monitor digitally
- **Bottom 10% Safe** 18.4% Sample Churn

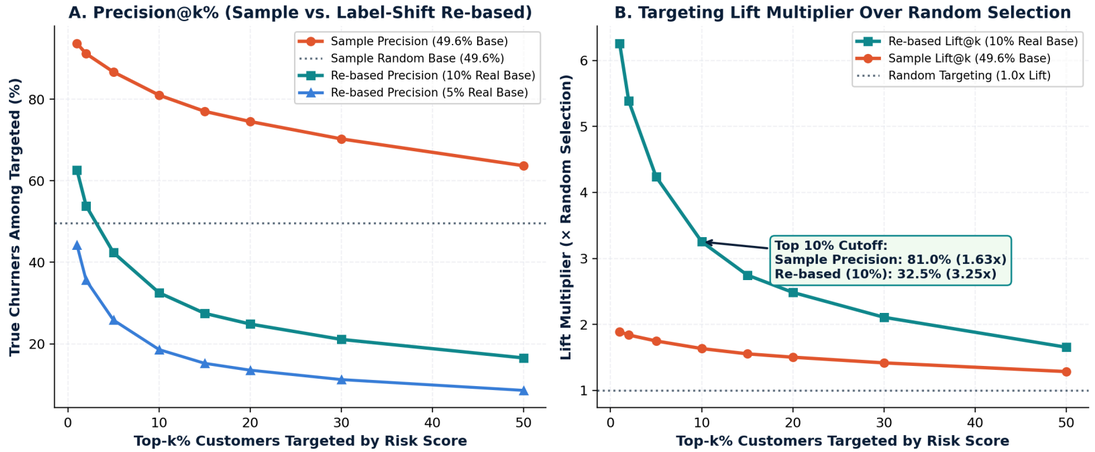
Triangulating Gain, Held-Out Permutation Importance & Ablation [10, 12]

Signal Family	Top Features (Gain Share / Perm Drop)	PDP / Ablation Finding	Mapped Retention Action
1. Handset Lifecycle	`dev_age_gt_300` (15.2% gain), `eqpdays` (4.6% gain, #3 perm), `refurb_new` (2.6%)	PDP jumps from **0.458 (300d) → 0.549 (315d)**; dropping all `eqpdays` cols cuts AUC by **-0.0065**.	**Lane 2:** Pre-300d trade-in voucher + quality warranty on refurb replacements.
2. Tenure Renewal Window	`months` (2.6% gain, #2 perm +0.0214), `tenure_le_9` (2.2%), `tenure_10_12_cliff` (1.9%)	PDP peaks at **0.609 at Month 12** before decaying to 0.475 by Month 24.	**Lane 2:** Proactive Month 8–9 renewal check before Month 11 peak.
3. Usage Slump & Plan Fit	`mou_ratio_3m` (#1 perm +0.0304, 2.3% gain), `asl_flag` (1.4%), `recurring_rev_share` (1.4%)	Users with `mou_Mean / avg3mou < 0.7` and high overage share spike in risk while still active.	**Lane 1:** Right-sized bundle + Account Spending Limit (`asl_flag`) activation.

So What / Decision Implication: A non-technical CRM manager can read every top model driver as a concrete action: **refresh aging handsets before Day 300** and **right-size plans when 3-month usage drops** [10, 13].

Sources: [10] Chen & Guestrin (2016); [12] Pedregosa et al. (2011) (`outputs/tables/feature_importance_gain.csv`, `feature_importance_permutation.csv`).

Top-10% Targeting Concentrates Churners by 3.25× (10% Prior) to 3.72× (5% Prior) Over Random [8, 9]



Held-Out Test Set Top-k% Targeting Table (Scaled to N = 100,000 Base; Sample vs. Label-Shift Re-Based) [8, 9]

Top-10% Risk (81.0% Sample Churn) vs. Bottom-10% Safe Tier (18.4% Sample Churn) = 4.40× Risk Separation

Targeting Tier (k%)	Targeted Customers (per 100k)	Score Cutoff ($\hat{p}_i$)	Sample Precision (π_s=49.6%)	Sample Recall (Share Caught)	Sample Lift	Re-Based Precision (π_r=10% Prior)	Re-Based Precision (π_r=5% Prior)
Top 1%	1,000	≥ 0.892	**93.7%**	1.9%	1.89×	**62.6%** (6.26×)	**44.2%** (8.84×)
Top 2%	2,000	≥ 0.866	**91.2%**	3.7%	1.84×	**53.8%** (5.38×)	**35.6%** (7.12×)
Top 5%	5,000	≥ 0.792	**86.7%**	8.7%	1.75×	**42.4%** (4.24×)	**25.8%** (5.17×)
Top 10% PILOT CUTOFF	10,000	≥ 0.713	**81.0%**	16.3%	1.63×	**32.5%** (3.25×)	**18.6%** (3.72×)
Top 15%	15,000	≥ 0.667	**77.0%**	23.3%	1.55×	**27.5%** (2.75×)	**15.2%** (3.04×)
Top 20%	20,000	≥ 0.635	**74.5%**	30.1%	1.50×	**24.9%** (2.48×)	**13.5%** (2.71×)
Top 30%	30,000	≥ 0.584	**70.3%**	42.5%	1.42×	**21.1%** (2.11×)	**11.2%** (2.25×)
Top 50%	50,000	≥ 0.497	**63.7%**	64.2%	1.28×	**16.5%** (1.65×)	**8.6%** (1.72×)

So What / Decision Implication: Selecting the **Top-10% cutoff ($\hat{p}_i \geq 0.713$, n = 10,000)** captures **3,253 true churners** under a 10% real base rate (vs. 1,000 by random outreach)—maximizing campaign efficiency [8, 9].

Sources: [8] Verbeke et al. (2012); [9] Saerens et al. (2002) (`outputs/tables/topk_targeting.csv`).

Operational Playbook: A Two-Lane Retention Program Gated by a 90-Day 80/20 Holdout Pilot [7, 8]



Lane 1: High-Value Usage-Slump & Plan-Fit Rescue

- **Target Segment:** Top-10% risk score (`p̂_i ≥ 0.713`) with active revenue (`rev_Mean ≥ 30 MU` , `mou_Mean > 0`) exhibiting a 3-month usage drop (`mou_ratio_3m < 0.75`) or overage friction (`ovrrev_Mean > 10 MU`).
- **Commercial Offer:** Proactive care call + right-sized voice/data bundle, one-time overage buffer credit, and Account Spending Limit (`asl_flag = Y`) activation.
- **Unit Economics:** Expected cost **c = 7.50 MU** per targeted account; protects **V = 328.83 MU** (8m margin at 58.72 MU ARPU). Break-even save rate is just **s* = 7.01%**.

Lane 2: Pre-300-Day Handset Refresh & Month 8–9 Check

- **Target Segment:** Active subscribers in the **240–300 day handset window** (or Tenure Months 8–9) with `rev_Mean ≥ 30 MU` (**8,815 accounts** in 100k; mean ARPU = **66.91 MU**, V = 374.70 MU).
- **Commercial Offer:** Early handset trade-in / 12-month installment upgrade voucher + quality check for refurbished-device holders (`refurb_new == 'R'`).
- **90-Day Pilot Math:** 80% treatment (`7,052` accts) vs. 20% control (`1,763` accts) at **c = 4.00 MU**; closing 25% of the 5.5 pp re-based jump saves **97.0 accounts** = **+8,125 MU net pilot return**.

Guardrail Lane: Automated Digital Triage for Dormant SIMs

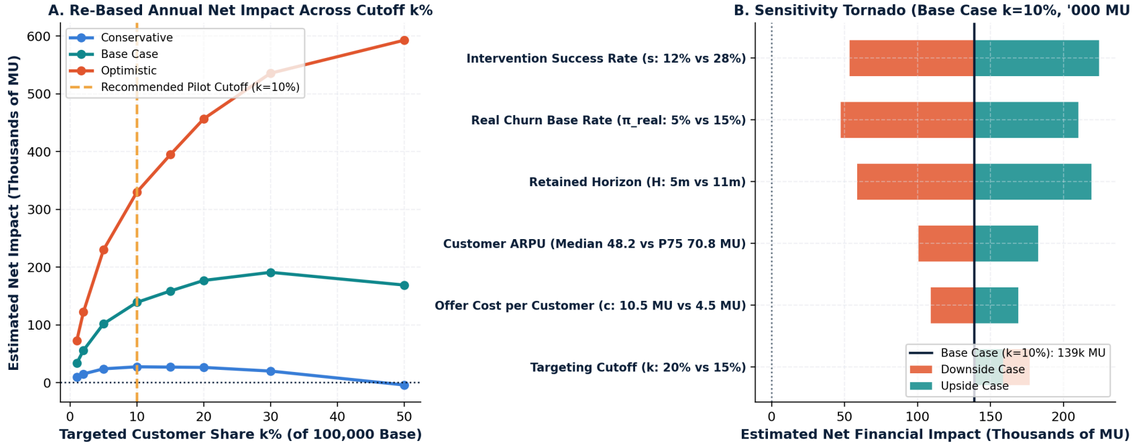
- **Target Segment:** High-risk accounts with `mou_Mean ≤ 10` or `rev_Mean < 30 MU` (including the 1,615 zero-MOU accounts with 79.81% churn and 23.11 MU ARPU).
- **Why Exclude From Human Calls:** At 23.11 MU ARPU, human call + subsidy outreach has poor unit economics because disengagement has already occurred.
- **Automated Action:** Zero-marginal-cost automated SMS/push "Win-Back Top-Up Bonus" (`c < 0.25 MU`) triggered only upon an inbound recharge or usage event.

So What / Decision Implication: Routing high-risk subscribers into **cause-specific lanes** and keeping a **mandatory 20% randomized holdout** turns predictive scores into verifiable profit [7, 8].

Sources: [7] Neslin et al. (2006); [8] Verbeke et al. (2012); [13] Company A Dataset (`docs/assumptions.md`).

6. ESTIMATED FINANCIAL IMPACT & SENSITIVITY

Re-Based Net Impact at Top-10%: +27,418 MU (Conservative) to +138,938 MU (Base) to +329,786 MU (Optimistic) [8, 9]



Explicit Scenario & Assumption Table at Pilot Cutoff k = 10% (n = 10,000 Targeted out of N = 100,000; Dataset Monetary Units, MU) [8, 9, 13]

Net Impact = n · p_adj · s · (ARPU · margin · H) − n · c

Scenario Name	Real Churn Prior (π_r)	Re-Based Precision (p_{adj})	Save Rate (s)	ARPU × Margin × H = V	Offer Cost (c)	Saved Accts	Gross Saved (MU)	Total Cost (MU)	Estimated Net Impact (MU)	ROI / Break-Even s*
Conservative (5% Prior, Median ARPU)	5.0%	18.59%	15.0%	48.20 × 65% × 6m = **187.98 MU**	2.50 MU	278.8	52,418 MU	25,000 MU	**+27,418 MU**	+109.7% / s*=7.15%
Base Case (Recommended) (10% Prior, Mean ARPU)	**10.0%**	**32.53%**	**20.0%**	58.72 × 70% × 8m = **328.83 MU**	**7.50 MU**	**650.6**	**213,938 MU**	**75,000 MU**	**+138,938 MU**	**+185.3% / s*=7.01%**
Optimistic (10% Prior, 12m Full Renewal)	10.0%	32.53%	25.0%	58.72 × 75% × 12m = **528.48 MU**	10.00 MU	813.2	429,786 MU	100,000 MU	**+329,786 MU**	+329.8% / s*=5.82%
Unadjusted Sample Reference (If 49.56% Prior Ignored)	*49.56%*	*81.00%*	*20.0%*	*58.72 × 70% × 8m = 328.83 MU*	*7.50 MU*	*1,620.0*	*532,708 MU*	*75,000 MU*	*+457,708 MU (3.3× inflated)*	*+610.3% / s*=2.82%*

So What / Decision Implication: Even under the **Conservative 5% real-churn prior (+27,418 MU net)** and downside tornado stress tests, the Top-10% program remains solidly profitable because break-even conversion is only **7.01%–7.15%** [8, 9].

Sources: [6] Reichheld & Sasser (1990); [8] Verbeke et al. (2012); [9] Saerens et al. (2002) (`outputs/tables/sensitivity_tornado.csv`, `docs/assumptions.md`).

Five Operational Risk Mitigations & a 90-Day Gated Pilot Roadmap Before Full Scale-Up [8, 9]

Key Technical & Commercial Risks Paired with Concrete Mitigations [8, 9, 13]

Risk / Limitation	Why It Matters	Concrete Mitigation Built Into PoC
1. Observational vs. Causal Uplift	High churn risk $\hat{p}_i$ does not guarantee a customer will accept an offer (*persuadables* vs. *sure losses*).	Enforce a **20% randomized holdout control group** in Phase 2 to measure true incremental save rate *s* before scaling budget.
2. Case-Control Prior Drift	Model probabilities are calibrated to the 49.56% sample prior rather than live monthly churn (5%–10%).	Apply **Saerens et al. [9] prior-shift adjustment** in batch scoring and monitor monthly Brier/decile calibration.
3. "Sleeping Dog" Trigger Risk	Contacting satisfied month-to-month customers about contract renewal can accidentally prompt shopping.	Restrict proactive calls to **Top-10% risk (** `p̂_i ≥ 0.713` **)**; track weekly treatment-vs-control churn delta by segment.
4. Subsidy Cannibalization	Low-risk or low-ARPU subscribers may game handset trade-in vouchers.	Enforce `rev_Mean ≥ 30 MU` **floor**, `mou_Mean > 0`, and 12-month installment credit vesting (no upfront cash).
5. Demographic Proxy Bias	Neighborhood/credit codes (`prizm_social_one`, `crclscod`) could correlate with protected groups.	`ethnic` **is excluded** (`-0.0016` AUC impact); audit offer eligibility parity across regions every 30 days.

90-Day Phased Implementation & Governance Roadmap

- **Phase 1 — Days 1 to 30: Shadow Scoring & Pipeline Integration**
  Deploy `src/features.py` (107 fair features) + `final_model.joblib` (M5) in monthly shadow mode on billing warehouse; validate data drift, missingness flags (`change_mou`, `hnd_webcap`), and CRM call-queue routing.
- **Phase 2 — Days 31 to 60: Launch 80/20 Randomized Holdout Pilot**
  Execute **Lane 1** (Top-10% usage-slump rescue: 8,000 treatment / 2,000 control) and **Lane 2** (`eqpdays` 240–300d refresh: 7,052 treatment / 1,763 control). Track acceptance rate, call handling time, and 30/60-day survival.
- **Phase 3 — Days 61 to 90: Uplift Readout, Uplift Model & Scale Gate**
  Compare 90-day churn between 80% Treatment and 20% Control. **Hard Scale Gate:** Proceed to full base rollout if verified save rate exceeds break-even (**s ≥ 7.01%** in Lane 1; net positive MU in Lane 2), then train a two-model uplift ranker on pilot outcomes.

So What / Decision Implication: Leadership does not need to gamble on unproven assumptions—a **90-day 80/20 pilot** validates whether conversion exceeds the **7.01% break-even threshold** before full commitment [8].

Sources: [8] Verbeke et al. (2012); [9] Saerens et al. (2002); `docs/problem_definition.md`, `docs/assumptions.md`.

8. VERIFIED REFERENCES & AI DISCLOSURE

Complete Numbered References & Mandatory Tool Disclosure (100% Matched to `references.txt`)

All External Market Reports, Peer-Reviewed Papers, Software Libraries, Dataset & AI Assistance Links [1]–[14]

Synchronized with `submission/references.txt`

[1] DataReportal (Kemp, S.). (2025). Digital 2025: Iraq & Digital 2024: Iraq — Global Digital Insights (46.0M–48.1M cellular mobile connections; 100%–103% population penetration). https://datareportal.com/reports/digital-2025-iraq

[2] World Bank & International Telecommunication Union (ITU). (2024). World Development Indicators: Mobile cellular subscriptions (per 100 people) — Iraq (IT.CEL.SETS.P2: 100.08 per 100 people). https://data.worldbank.org/indicator/IT.CEL.SETS.P2?locations=IQ

[3] GSMA Intelligence. (2024). The Mobile Economy Middle East and North Africa 2024 (427M unique MENA subscribers; 64% penetration; prepaid dominance). https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-economy/mena/

[4] Zain Group. (2024). Zain Group Annual Report 2023 & 2024 Operations Review — Zain Iraq (17.9M–19.3M active subscribers; USD 0.97B–1.1B revenue). https://www.zain.com/en/investor-relations/financial-reports

[5] Ooredoo Group. (2024). Ooredoo Group Annual Report 2023 & 2024 — Asiacell Iraq Operations (17.69M–19.13M subscribers; QAR 4,452M revenue; 24.8 QAR blended ARPU). https://www.ooredoo.com/en/investors/financial-information/annual-reports/

[6] Reichheld, F. F., & Sasser, W. E., Jr. (1990). Zero Defections: Quality Comes to Services. Harvard Business Review, 68(5), 105–111 (5% defection reduction yields 25%–85% profit lift). https://hbr.org/1990/09/zero-defections-quality-comes-to-services

[7] Neslin, S. A., Gupta, S., Kamakura, W., Lu, J., & Mason, C. H. (2006). Defection Detection: Measuring and Understanding the Predictive Accuracy of Customer Churn Models. Journal of Marketing Research, 43(2), 204–211. https://doi.org/10.1509/jmkr.43.2.204

[8] Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. European Journal of Operational Research, 218(1), 211–229. https://doi.org/10.1016/j.ejor.2011.09.031

[9] Saerens, M., Latinne, P., & Decaestecker, C. (2002). Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure. Neural Computation, 14(1), 21–41. https://doi.org/10.1162/089976602753284446

[10] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16), 785–794. https://doi.org/10.1145/2939672.2939785

[11] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Advances in Neural Information Processing Systems (NeurIPS 2017), 30, 3149–3157. https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[12] Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

[13] Global Consumer Intelligence (GCI) & Teradata Center at Duke University. (2026). GCI World 2026 September Final Assignment: Company A Telecom Dataset (Client.csv, Record.csv) and Dataset Overview. https://edu.omnicamp.us/en/courses/170/reports/125/

[14] AI Assistance Disclosure (Google Antigravity / Gemini). (2026). AI Ideation, Literature Search, Code Verification & Slide Layout Assistance Log — Shared Conversation Link: [REPLACE_WITH_SHARED_AI_CHAT_URL_BEFORE_SUBMISSION]

Reproducibility & Auditability Note: Every metric in Slides 1–14 is programmatically verified against `outputs/results.json` generated by `submission/ACCOUNT_NAME.ipynb` (`SEED = 42`, 0 mismatches).